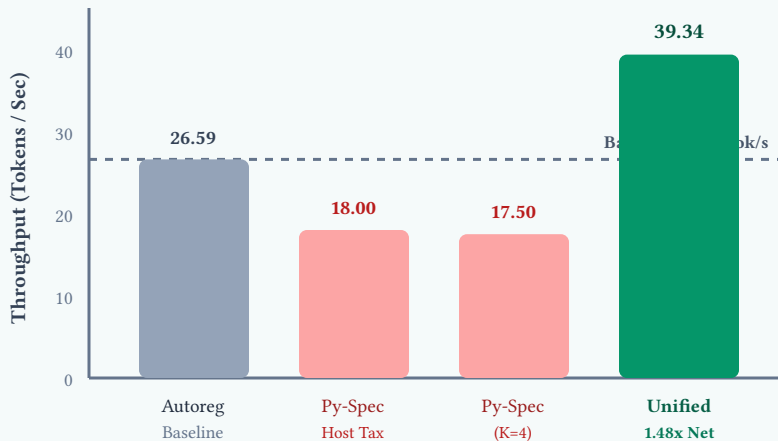


UNIFIED SPECULATIVE SERVING ENGINE & STATIC KV-CACHE PERFORMANCE

Physical T4 Throughput (tok/s) and Static KV-Cache Microsecond Rollback Speedup (Table 5)

Panel A: Serving Throughput (Tokens / Sec)



Panel B: Static KV-Cache Microbenchmarks

Rollback Latency (drop_k = 3)

Dynamic torch.cat:	7353.6 μ s	
Static KV-Cache (Ours):	0.51 μ s	(14,388x faster)

DRAM Memory Traffic per Token

Dynamic Reallocation:	120,344 KB / tok	
Static KV-Cache (Ours):	56 KB / tok	(2,149x cut)

Pre-allocated tensor pointers guarantee pointer invariance